

Trapezoidal rule for integration

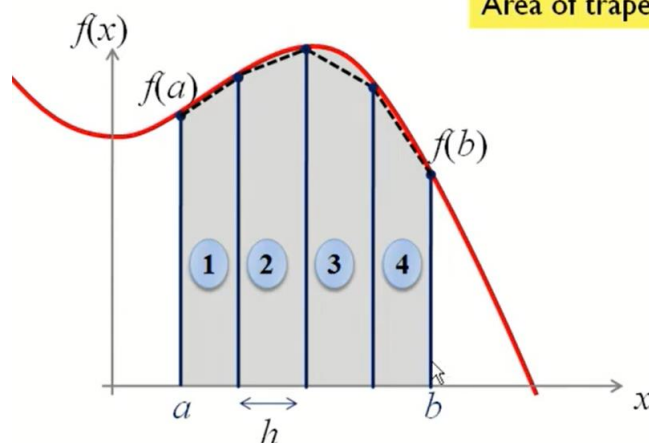
Numerical Integration

► The Trapezoidal Rule

► How does it work?

$$I = \int_a^b f(x) dx$$

Area of trapezoid = height/2 × (base 1 + base 2)



$$I_1 = \frac{h}{2}(f(a) + f(a+h))$$

$$I_2 = \frac{h}{2}(f(a+h) + f(a+2h))$$

$$I_3 = \frac{h}{2}(f(a+2h) + f(a+3h))$$

$$I_4 = \frac{h}{2}(f(a+3h) + f(b))$$

Trapezoidal rule:

It is a numerical method we used to approximate the definite integral of our function. It estimates the area under a curve by dividing it into trapezoids instead of rectangles

The way it work is that if we have an interval $[a,b]$, we divide it into small equal parts which is here (h) , and use the area under the curve bounded by trapezoid, this may be cause an error , so it is better to divide the interval into many small equal part to get a correct approximation.

Numerical Integration

► The Trapezoidal Rule

► How does it work?

$$I = \int_a^b f(x) dx$$

$$I \approx \frac{h}{2} \left(f(a) + f(b) + 2 \sum_{k=1}^{n-1} f(a + kh) \right)$$

$$h = \frac{b-a}{n}$$

$n \rightarrow$ number of segments
 $h \rightarrow$ step size

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This is the general rule,

$$I \approx \frac{h}{2} (f(a) + 2f(a+h) + 2f(a+2h) + 2f(a+3h) + f(b))$$

IMPORTANT NOTE HERE: The first and last terms are multiplied by 1, while all inner points are multiplied by 2 because they are shared by two adjacent trapezoids.

Application in MATHS,ENGINEERING :

- 1-Numerical approximation of definite integrals when exact integration is difficult
- 2-Used in solving differential equations numerically
- 3-Calculating area under curves from experimental data point

Matrix normalization

We can call it as a process of scaling the values of a matrix to a specific range or distribution from [0 to 1].

It is commonly used to transform data so that all values become comparable and stable for numerical algorithms.

There are multiple types of matrix normalization :

1- min & max:

This method scales the elements of the matrix to a specific range, usually [0, 1] .

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Score Normalization (Standardization)

This method centers the matrix around the mean and scales according to the standard deviation:

$$a_{ij}^{\text{standardized}} = \frac{a_{ij} - \mu}{\sigma}$$

Where μ is the mean of all elements and σ is the standard deviation. This helps in bringing the data to have zero mean and unit variance.

Applications of Matrix Normalization in AI

Neural Networks Training

Neural networks perform better when inputs are normalized.

Without normalization:

Gradients can explode or vanish

Training becomes slower and unstable

With normalization:

Faster convergence

More stable
learning

Example:

Image pixels range: $0 \rightarrow 255$

We normalize to: $0 \rightarrow 1$ before feeding into a neural network.